nanoker

CERAMIC TECHNICAL DATA SHEET

KA998

Alumina (Al_2O_3) 99.85%

KEY PROPERTIES

- Excellent abrasion resistance
- Excellent dielectric properties
- High corrosion resistance
- Good thermal conductivity

APPLICATIONS

- Labware
- Furnace construction
- Abrasives industry
- Machinery parts
- Ballistic protection
- Chemical industry

MAIN PROPERTIES

Property	Value	Unit
Density	3.9	g/cm³
Flexural Strength	350	MPa
Hardness	16	GPa
Fracture Toughness	3.5	MPa·√m
Young Modulus	370	GPa
Thermal Conductivity (100°C)	30	W/m·K
CTE (20–1000ºC)	8.6	$x10^{-6}$ K^{-1}

** All properties measured at 20ºC unless otherwise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1550	ºC
Recomended temp	<1500	ºC
Continuous temp	1400	ºC

ELECTRICAL BEHAVIOR

Resistivity (25ºC)	>1E14	Ω·cm
Dielectric strength	14	kV/mm

MATERIAL INFORMATION

Material type	Al_2O_3
Machinability	Medium
Structure	Dense ceramic
Purity	99.85%
Color	Ivory

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

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